IBM Fusion

Fusion Access for SAN

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IBM Fusion Access for SAN: Overview

Introduction

IBM Fusion Access for SAN (powered by **IBM Storage Scale Container Native Storage Access (CNSA)**) provides high-performance, low-latency block storage access for Red Hat OpenShift Virtualization workloads over Fibre Channel (FC) or iSCSI protocols.

Product Disclaimer

IBM Fusion is being developed and released in an agile manner. In addition to adding new capabilities, the interface is likely to change over time. Therefore, the screenshots used in this document and other lab guides may not always look exactly like what you see in the product. Depending on timing, you can expect to encounter some of the following:

- Changes in the user interface (UI), such as the location of buttons or text in various fields
- Additional tabs or buttons

These differences should not affect how the lab exercises work. When changes are observed, have patience and explore.

Getting Help

If you require assistance in interpreting any of the steps in this guide, please post your questions to the #storage_demo_feedback Slack channel for IBMers only. Business Partners can request help at the Partner Plus Support website.

For troubleshooting tips, see the TechZone Set Up Troubleshooting Guide for help with common issues and solutions when using IBM Technology Zone. Additionally, go to the IBM Technology Zone Help page. If you have an issue with the site, you can open a support case. Alternatively, you can send an email to techzone.help@ibm.com.

For help with the Fusion product, see the Fusion documentation.

Prerequisites & Getting Started

You **must** be familiar with the Linux operating system, as well as know how to use the command line interface to execute basic Linux commands such as **kubectl**, **oc**, **exit**, **ls**, **mkdir**, **ssh**, **sudo**, and **curl**. Ideally, you should also have some experience working with the Red Hat OpenShift Graphical User

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Interface (GUI) and executing Red Hat OpenShift Container Platform and Kubernetes command line interface (CLI) commands.

When instructions are given to enter text, the text is usually shown in mono space font `this is an example`. This is to help indicate that you need to type or copy/paste all the text as is.

For detailed step-by-step instructions on creating an IBMid and requesting your TechZone environment, see the TechZone Quick Start Guide.

Key Capabilities

- **Easy User Experience:** Wizard-driven user interface to install and configure storage clusters, file systems, and storage classes.
- **Use Existing Infrastructure:** Leverage existing SAN investments, including Fibre Channel (FC) and iSCSI technologies, as you transition to or expand with OpenShift Virtualization.
- **Scalability:** Storage cluster designed to scale with OpenShift Container Platform clusters and virtual machine workloads. Supports up to approximately 3000 VMs on 6 bare-metal hosts.
- **Consolidated and Shared Storage:** Multiple servers access large, shared data storage capacity with automatic data backup and continuous monitoring.
- **High-Speed Data Transfer:** Dedicated high-speed network for storage overcomes data transfer bottlenecks of traditional LAN.
- **File-Level Access:** File systems built on top of SAN storage provide file-level access through shared-disk file systems.
- **Centralized Management:** Centralized management and configuration of SAN components like Logical Unit Numbers (LUNs).

Architecture

IBM Fusion Access for SAN is based on **IBM Storage Scale Container Native Storage Access (CNSA)** using local filesystems. The architecture uses IBM Storage Scale as a containerized clustered file system that provides concurrent access from multiple nodes.

IBM Storage Scale CNSA deploys as: * **Core pods** on selected worker nodes (storage nodes) * **CSI driver** with operator, CSI sidecars, and CSI node plugins * **Local filesystems** backed by shared LUNs accessible by all worker nodes simultaneously

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Use Cases

- High-performance, low-latency storage for OpenShift Virtualization VMs
- Migrating VMware workloads to OpenShift Virtualization with comparable storage performance
- Enterprise storage abstraction independent of underlying storage vendor
- OpenShift-native user interface for storage management

References

- IBM Fusion Access for SAN documentation: IBM Fusion Access for SAN
- IBM Storage Scale CNSA overview: IBM Storage Scale CNSA
- IBM Storage Scale CSI driver: IBM Storage Scale CSI Driver
- Video overview: IBM Fusion Access for SAN Video

Resources

- IBM Fusion Collection on TechZone: IBM Fusion Collection
- IBM Fusion Environments: Environments
- IBM Fusion Resources: Resources

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IBM Fusion Access for SAN: Prerequisites & Limitations

Prerequisites

Before installing and configuring IBM Fusion Access for SAN, ensure the following requirements are met:

Infrastructure Requirements

- **Bare-metal worker nodes** with attached SAN storage
- **A working container registry** enabled
- **All worker nodes must connect to the same LUNs** — a shared LUN is a shared disk accessed by all worker nodes simultaneously
- **Kubernetes pull secret** for the IBM Cloud Container Registry (ICR)

OpenShift Cluster Requirements

- Red Hat OpenShift 4.21 or later
- IBM Fusion 2.14 or later (GA since 2.11)
- Worker nodes: minimum 3 dedicated worker nodes for storage cluster
- OpenShift cluster must have the IBM Operator Catalog configured
- Entitlement key for IBM Cloud Container Registry

Limitations

IBM Fusion Access for SAN Limitations

- **Hosted Control Plane (HCP) clusters are not supported**

IBM Storage Scale CNSA Limitations (applicable to Fusion Access for SAN)

Since IBM Fusion Access for SAN is based on IBM Storage Scale Container Native Storage Access (CNSA), the following CNSA limitations apply:

- Deployment of IBM Storage Scale pods on master nodes is not supported except for "compact" Red Hat OpenShift clusters
- Deployment of IBM Storage Scale pods on nodes with ARM CPUs is not supported

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- Single node IBM Storage Scale container native clusters are not supported
- Maximum number of worker nodes supported: **128**
- Maximum number of local filesystems supported: **3**
- Maximum number of storage nodes: **32**
- Maximum number of disks per local filesystem: **128**

References

- Prerequisites documentation: IBM Fusion Access for SAN Prerequisites

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Lab Environment Overview

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

TechZone Reservation Configuration

Reserve the **TechZone Fusion Environment** with the following configuration:

- **OpenShift Version:** 4.21
- **Worker Nodes:** 3 dedicated worker nodes (plus infrastructure nodes)
- **Worker Node Size:** 16 or 32 Core Worker Nodes

Cluster Verification

Once the cluster is provisioned, verify the environment:

```
export KUBECONFIG=~/kubeconfig
oc get no
```

Expected output showing master, infra, and worker nodes:

```
NAME                 STATUS   ROLES                  AGE   VERSION
itz-h21pns-infra-1   Ready    worker                 8h    v1.34.9
itz-h21pns-infra-2   Ready    worker                 8h    v1.34.9
itz-h21pns-infra-3   Ready    worker                 8h    v1.34.9
itz-h21pns-master-1  Ready    control-plane,master   8h    v1.34.9
itz-h21pns-master-2  Ready    control-plane,master   8h    v1.34.9
itz-h21pns-master-3  Ready    control-plane,master   8h    v1.34.9
itz-h21pns-worker-1  Ready    worker                 8h    v1.34.9
itz-h21pns-worker-2  Ready    worker                 8h    v1.34.9
itz-h21pns-worker-3  Ready    worker                 8h    v1.34.9
itz-h21pns-worker-4  Ready    worker                 8h    v1.34.9
```

Verify cluster version:

```
oc get clusterversion
```


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IBM Storage Scale Node

The lab environment includes a single-node IBM Storage Scale cluster:

- **Version:** 6.0.0.1
- **NSDs:** 2 NSDs configured, each 2TB in size
- **Filesystem:** `fs1` created using both NSDs (4TB total capacity)
- **IP Address:** `10.10.10.202`
- **Network:** Single 10GbE interface

Cluster Preparation

Required Machine Configurations

Apply the following Machine Configurations (MCO) to prepare the OpenShift cluster:

1. iSCSI Multipath MCO

The MachineConfig to enable `iscsid`, `iscsi`, and `multipathd` services is available at `examples/yaml/iscsi-mco.yaml`:

```
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: worker
  name: 99-worker-iscsi-configuration
spec:
  config:
    ignition:
      version: 3.2.0
    systemd:
      units:
        - name: iscsid.service
          enabled: true
        - name: iscsi.service
          enabled: true
        - name: multipathd.service
          enabled: true
    osImageURL: ""
```

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Apply and monitor:

```
oc apply -f iscsi-mco.yaml
oc get mcp
```

Wait until the worker MCP shows `UPDATED True` and `UPDATING False`.

2. OpenShift Kernel Development MCO

From the bastion or CLI:

```
oc apply -f https://raw.githubusercontent.com/IBM/ibm-spectrum-scale-container-native/v6.0.0.x/generated/scale/mco/mco.yaml
oc get mcp
```

Wait for completion.

3. Install the "kernel module management" operator

Install via OperatorHub or CLI as needed for the Scale module build.

Scale Node Access

Login to the Scale node:

```
ssh -i {ssh-key-path} {scale-ssh-user}@{scale-node-ip}
sudo -i
```

Verify the filesystem:

```
df -h {scale-fs-mount}
```

Expected output (using defaults; adjust for your environment):

```
Filesystem      Size  Used Avail Use% Mounted on
{scale-fs-name}            {scale-fs-size}  5.2G  995G   1% {scale-fs-mount}
```

This filesystem will be used as the backing store for the Linux-IO iSCSI target LUNs.

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Setting up iSCSI using Linux-IO (LIO)

Introduction

This section covers configuring iSCSI on OpenShift worker nodes using Linux-IO (LIO) as the iSCSI target, with IBM Storage Scale as the backing store.

About iSCSI

A little-known fact: IBM together with Cisco Systems jointly developed the iSCSI protocol in 1998. It was submitted as a draft standard in 2000 and officially published as RFC 3720 by the IETF in 2004.

iSCSI encapsulates SCSI commands and data into TCP/IP packets for transmission over a standard network. It is widely used due to its cost-effectiveness and simplicity compared to alternatives like Fibre Channel.

The 3 core components of iSCSI: * **iSCSI initiator:** Client-side software/hardware that connects to storage * **iSCSI target:** Storage device that receives connections and serves storage * **iSCSI Qualified Name (IQN):** Unique name used to address iSCSI targets

Why iSCSI for OpenShift?

In the context of OpenShift, deploying worker nodes with Fibre Channel adapters connected to SAN-attached storage is costly. iSCSI is a more efficient and cost-effective alternative. We use the Linux-IO (LIO) target, which is included in the Linux kernel and supports a wide variety of protocols including FCoE and iSCSI.

Example Scripts

The complete configuration scripts used in this guide are available in the repository examples directory:

- `examples/scripts/iscsi-init.sh` — iSCSI initiator setup on worker nodes
- `examples/scripts/targetcli-config.sh` — iSCSI target configuration on Scale node
- `examples/yaml/iscsi-mco.yaml` — MachineConfig for iSCSI/multipath enablement
- `examples/yaml/targetcli-saveconfig.json` — Reference targetcli saveconfig output

Step 1: Prepare the Scale Node

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Install Required Packages

```
yum install targetcli
sudo dnf install iscsi-initiator-utils -y
```

Configure Firewall

Open port 3260 for iSCSI:

```
firewall-cmd --permanent --add-port {iscsi-target-port}/tcp
systemctl stop firewalld
systemctl start firewalld
firewall-cmd --list-ports
```

Expected output (port may vary):

```
1191/tcp {iscsi-target-port}/tcp 60000-61000/tcp
```

Verify Filesystem

```
df -h {scale-fs-mount}
```

This filesystem (`fs1`) will be used as the backing store for iSCSI LUNs.

Step 2: Configure iSCSI Initiator on Worker Nodes

Apply iSCSI Multipath MachineConfig

Apply the MachineConfig to enable `iscsid`, `iscsi`, and `multipathd` services on all worker nodes:

```
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: worker
  name: 99-worker-iscsi-configuration
spec:
  config:
    ignition:
      version: 3.2.0
    systemd:
      units:
        - name: iscsid.service
```


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```
          enabled: true
        - name: iscsi.service
          enabled: true
        - name: multipathd.service
          enabled: true
    osImageURL: ""
```

```
oc apply -f iscsi-mco.yaml
oc get mcp
```

Wait until the worker MCP shows `UPDATED True` and `UPDATING False`.

Enable Multipath and iSCSI on Worker Nodes

The complete initiator setup is automated in `examples/scripts/iscsi-init.sh`. Key steps:

```
#!/bin/bash
# iSCSI Initiator Configuration Script for OpenShift Worker Nodes
# Run from bastion with oc CLI configured
#
# Environment variables (with defaults matching fusion-access-san-attrs.adoc):
#   SSH_KEY_PATH       - SSH private key path (default: /tmp/ocp/cluster/id_rsa)
#   BASTION_USER       - SSH user for worker nodes (default: core)
#   WORKER_NODE_PREFIX - Worker node name prefix for filtering (default: worker-)

set -e

: "${SSH_KEY_PATH:=/tmp/ocp/cluster/id_rsa}"
: "${BASTION_USER:=core}"
: "${WORKER_NODE_PREFIX:=worker-}"

echo "=== iSCSI Initiator Configuration for Worker Nodes ==="
echo "Using SSH_KEY_PATH=${SSH_KEY_PATH}, BASTION_USER=${BASTION_USER},
WORKER_NODE_PREFIX=${WORKER_NODE_PREFIX}"

# 1. Enable multipath on all worker nodes
echo "Enabling multipath on worker nodes..."
for WORKER in $(oc get nodes | grep "${WORKER_NODE_PREFIX}" | awk '{print $1}'); do
  echo "  Configuring $WORKER..."
  ssh -i "${SSH_KEY_PATH}" "${BASTION_USER}@$WORKER" sudo mpathconf --enable
done

# 2. Enable and start iSCSI service on all worker nodes
```

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```
echo "Enabling iSCSI service on worker nodes..."
for WORKER in $(oc get nodes | grep "${WORKER_NODE_PREFIX}" | awk '{print $1}'); do
  echo "  Enabling iscsi on $WORKER..."
  ssh -i "${SSH_KEY_PATH}" "${BASTION_USER}@$WORKER" sudo systemctl enable iscsi
done

for WORKER in $(oc get nodes | grep "${WORKER_NODE_PREFIX}" | awk '{print $1}'); do
  echo "  Starting iscsi on $WORKER..."
  ssh -i "${SSH_KEY_PATH}" "${BASTION_USER}@$WORKER" sudo systemctl start iscsi
done

# 3. Get IQNs from worker nodes
echo "Retrieving IQNs from worker nodes..."
for WORKER in $(oc get nodes | grep "${WORKER_NODE_PREFIX}" | awk '{print $1}'); do
  echo "  $WORKER IQN:"
  ssh -i "${SSH_KEY_PATH}" "${BASTION_USER}@$WORKER" sudo cat
/etc/iscsi/initiatorname.iscsi
done

echo "=== iSCSI Initiator Configuration Complete ==="
echo "Next: Configure iSCSI target on Scale node using targetcli"
```

Obtain IQNs

Run the IQN retrieval on each worker node to get their IQNs for ACL configuration on the target.

```
for WORKER in $(oc get nodes | grep worker | awk '{print $1}'); do
  echo "$WORKER IQN:"
  ssh -i /tmp/ocp/cluster/id_rsa core@$WORKER sudo cat /etc/iscsi/initiatorname.iscsi
done
```

Step 3: Configure iSCSI Target on Scale Node using targetcli

`targetcli` uses a tree-like, filesystem-based structure. Navigate and make changes by moving through directories.

Configuration Flow Overview

1. Create backing store files in `backstore/fileio`
2. Create iSCSI target IQN in `iscsi/`

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3. Configure target portal group in `iscsi/<iqn>/tpg1/`
4. Create LUNs in `iscsi/<iqn>/tpg1/luns/`
5. Create ACLs for initiators in `iscsi/<iqn>/tpg1/acls/`
6. Save configuration

Create Backing Store Files

Navigate to backstore/fileio and create two 500G files:

```
targetcli
/backstores/fileio create iscsilun0 {scale-fs-mount}/iscsilun0 {lun-size}
/backstores/fileio create iscsilun1 {scale-fs-mount}/iscsilun1 {lun-size}
```

Verify creation:

```
/backstores/fileio ls
```

Create iSCSI Target IQN

```
/iscsi create {iscsi-target-iqn}
```

Configure Target Portal Group (TPG)

Navigate to the TPG directory (auto-created as `tpg1`):

```
/iscsi/{iscsi-target-iqn}/tpg1
```

Accept defaults: * Interface: `0.0.0.0` (listen on all interfaces) * Port: `3260`

Create LUNs

```
/iscsi/{iscsi-target-iqn}/tpg1/luns create /backstores/fileio/iscsilun0
/iscsi/{iscsi-target-iqn}/tpg1/luns create /backstores/fileio/iscsilun1
```

Verify:

```
/iscsi/{iscsi-target-iqn}/tpg1/luns ls
```

Create ACLs for Worker Nodes

Navigate to ACLs and create entries for each worker node IQN (replace with actual IQNs from Step 2):

```
/iscsi/{iscsi-target-iqn}/tpg1/acls create iqn.1994-05.com.redhat:<worker1-iqn>
```

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```
/iscsi/{iscsi-target-iqn}/tpg1/acls create iqn.1994-05.com.redhat:<worker2-iqn>
/iscsi/{iscsi-target-iqn}/tpg1/acls create iqn.1994-05.com.redhat:<worker3-iqn>
```

Note: Auto-mapping is enabled by default, so the two LUNs will be mapped to each client. This satisfies the prerequisite that all nodes see the same LUNs.

Verify and Save Configuration

```
cd /
ls
saveconfig
exit
```

Configuration saved to `/etc/target/saveconfig.json`. A reference copy is available at `examples/yaml/targetcli-saveconfig.json`.

Set File Permissions

```
chmod 777 {scale-fs-mount}/iscsilun*
ls -al {scale-fs-mount}/iscsilun*
```

Verify Target Service

```
netstat -an | grep {iscsi-target-port}
```

Expected: `tcp 0 0 0.0.0.0:3260 0.0.0.0:* LISTEN`

Step 4: Verify iSCSI Connectivity from Worker Nodes

iSCSI Discovery

Run iSCSI discovery on each worker node:

```
for WORKER in $(oc get nodes | grep {worker-node-prefix} | awk '{print $1}'); do
  ssh -i {ssh-key-path} {bastion-user}@$WORKER sudo iscsiadm -m discovery -t sendtargets
-p {scale-node-ip}
done
```

Each should return the target IQN.

Login and Session Verification

```
for WORKER in $(oc get nodes | grep {worker-node-prefix} | awk '{print $1}'); do
  ssh -i {ssh-key-path} {bastion-user}@$WORKER sudo iscsiadm -m node -T {iscsi-target-iqn}
```


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```
-p {scale-node-ip} -l
done
```

Verify sessions on the target:

```
targetcli sessions
```

Expected output showing all three workers logged in.

Verify LUN Visibility

Check that each worker sees the 2 500G LUNs:

```
for WORKER in $(oc get nodes | grep {worker-node-prefix} | awk '{print $1}'); do
  ssh -i {ssh-key-path} {bastion-user}@$WORKER sudo lsblk
done
```

Multipath Check

Since we only have a single target portal, multipath will not attach (single path). If multiple Scale nodes were configured as target portals, multipath would provide redundancy.

```
for WORKER in $(oc get nodes | grep {worker-node-prefix} | awk '{print $1}'); do
  ssh -i {ssh-key-path} {bastion-user}@$WORKER sudo multipath -ll
done
```

Step 5: Validate Persistence Across Reboot

Test that iSCSI LUNs persist a worker node reboot:

```
# Cordon, drain, and reboot a worker node (replace worker-3 with actual node name)
oc adm cordon {worker-node-prefix}3
oc adm drain {worker-node-prefix}3 --ignore-daemonsets --delete-emptydir-data
# Reboot via cloud provider console or IPMI
```

After reboot and node returns to Ready state:

```
oc adm uncordon {worker-node-prefix}3
# Verify iSCSI login and LUN visibility
for WORKER in $(oc get nodes | grep {worker-node-prefix} | awk '{print $1}'); do
  ssh -i {ssh-key-path} {bastion-user}@$WORKER sudo lsblk
done
```

Verify on Scale node:

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```
targetcli sessions
```

All 3 workers should show `LOGGED_IN` state.

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Deploying IBM Fusion Access for SAN

Introduction

This guide covers deploying IBM Fusion Access for SAN after the iSCSI infrastructure is configured. The procedure installs the IBM Fusion Operator, configures Fusion Data Foundation (FDF) for local storage, and enables the External Storage Area Network (SAN) service.

Official documentation: IBM Fusion Access for SAN Documentation

Note: The steps to add the IBM Operator Catalog and create the IBM entitlement key are already implemented in the TechZone cluster.

Step 1: Add IBM Operator Catalog (Reference)

The following is for reference only — already configured in TechZone:

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: ibm-operator-catalog
  namespace: openshift-marketplace
spec:
  sourceType: grpc
  image: icr.io/cpopen/ibm-operator-catalog:latest
  displayName: IBM Operator Catalog
  publisher: IBM
  updateStrategy:
    registryPoll:
      interval: 45m
```

Verify:

```
oc apply -f catalog_source.yaml -n openshift-marketplace
oc get CatalogSources ibm-operator-catalog -n openshift-marketplace
oc get pods -n openshift-marketplace
```

Step 2: Install IBM Fusion Operator

1. Log in to the OpenShift Web Console as `kubeadmin`.

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2. Navigate to Ecosystem › Software Catalog.
3. Search for `Fusion` and select the **IBM Storage Fusion** operator tile.
4. Click Install.
5. Leave all options as default:
    - **Update channel:** Stable channel (e.g., `v2.x`)
    - **Installation mode:** All namespaces on the cluster (default)
    - **Installed Namespace:** `ibm-spectrum-fusion-ns`
    - **Approval Strategy:** Automatic
6. Click Install.
7. Wait until Operator status shows **Succeeded** / **Installed operator - ready for use**.

Step 3: Connect to IBM Fusion Console

1. From the OpenShift Console top masthead, click the **Application Menu** (9-square grid).
2. Click **IBM Storage Fusion**. A new tab opens.
3. If prompted, select **kube:admin** and enter credentials.
4. Accept the license agreement: check **I have read and accept the license agreement**, click Continue.
5. You are redirected to the **Welcome to IBM Fusion** dashboard.

Step 4: Install Fusion Data Foundation (Local Storage)

Before configuring External SAN, install FDF for local storage on the cluster.

Prerequisite: Local Storage Operator must be installed.

1. In Fusion Console, navigate to **Storage → Local storage** and click **Get Started**.
2. Click **Create Storage Cluster**.
3. In OpenShift Console, **Create a new StorageClass using local storage devices** is pre-selected. Click Next.
4. Leave **Use Ceph RBD** as default StorageClass. Optionally select **Set default StorageClass for Virtualization**. Click Next.
5. Choose **Disks on selected nodes** and select the **infra nodes** for Data Foundation. Click Next.

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6. Select **Yes** to Create LocalVolumeSet. Click Next.
7. Use default options. Click Next.
8. Click **Create Storage systems**.
9. Wait for the storage cluster to become **Healthy**.

Validate Storage Classes

```
oc get storageclass
```

Create Storage for OpenShift Image Registry

1. Navigate to **Storage** → **PersistentVolumeClaims**.
2. Change to the `openshift-image-registry` namespace.
3. Create a new CephFS volume:
    - StorageClass: **CephFS**
    - Name: `imageregistry`
    - Size: `100G`
    - Access Mode: **RWX**
    - Click **Create**.
4. Patch the image registry:

```
oc patch configs.imageregistry.operator.openshift.io cluster \
  --type merge \
  -p '{"spec":{"managementState":"Managed","storage":{"pvc":{"claim":"imageregistry"}}}}'
```

1. Verify:

```
oc get clusteroperator image-registry
```

Step 5: Configure External SAN (Fusion Access for SAN)

Now configure the SAN storage using the iSCSI LUNs prepared earlier.

1. In Fusion Console, navigate to **Data Foundation** → **External Systems**.
2. Click **Connect external Systems**, select **Storage Area Network**, click Next.

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3. Switch from "All nodes" to **Select nodes**.
4. Select only the **worker nodes** (worker-1, worker-2, worker-3).
5. Name your filesystem: `scale1` (or `fusionaccessfs`).
6. Click **Connect and Create**.

Wait for the filesystem to become **Ready**.

Monitor Deployment

Monitor pods in the `ibm-spectrum-scale` namespace:

```
oc get pods -n ibm-spectrum-scale -w
```

Expected pods when ready:

```
NAME                              READY   STATUS    RESTARTS   AGE
ibm-spectrum-scale-gui-0          4/4     Running   0          13m
ibm-spectrum-scale-gui-1          4/4     Running   0          9m26s
ibm-spectrum-scale-pmcollector-0  2/2     Running   0          13m
ibm-spectrum-scale-pmcollector-1  2/2     Running   0          10m
worker-1                          2/2     Running   0          13m
worker-2                          2/2     Running   0          13m
worker-3                          2/2     Running   0          13m
```

Note: Unlike normal CNSA where core pods run on all worker nodes, for Fusion Access for SAN, core pods are only deployed on the selected worker nodes.

Check Custom Resources

```
oc get cluster -n ibm-spectrum-scale
oc get localdisk -n ibm-spectrum-scale
oc get filesystem -n ibm-spectrum-scale
```

Expected filesystem status progression:

```
NAME                  ESTABLISHED   HEALTHY       AGE
{storage-class-name}  False         NotReported   104s
...
{storage-class-name}  True          Healthy       3m26s
```

Access IBM Storage Scale CNSA GUI

```
echo -e "https://`kubectl get route -n ibm-spectrum-scale ibm-spectrum-scale-gui -ojson |
```


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```
jq -r .spec.host`"
```

Authenticate with OpenShift credentials to access the CNSA GUI dashboard.

Step 6: Verify Storage Class Creation

The Fusion Access for SAN operator automatically creates a StorageClass named `fusionaccessfs` (or your custom name).

```
oc get storageclass {storage-class-name} -o yaml
```

This StorageClass is used for provisioning RWX PVCs for OpenShift Virtualization VMs.

Filesystem Block Size

Note: The default filesystem block size is 4MB.

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Deploying a Virtual Machine using Fusion Access for SAN

Introduction

This guide demonstrates deploying a virtual machine using the IBM Fusion Access for SAN storage class (`fusionaccessfs`) with OpenShift Virtualization.

Important Fusion Access for SAN is only supported on bare-metal worker nodes. This lab environment uses nested virtualization for demonstration purposes only.

Step 1: Install OpenShift Virtualization Operator

1. Navigate to Ecosystem › Software Catalog (OperatorHub).
2. Search for **OpenShift Virtualization**.
3. Click on the operator to install it.
4. Leave all defaults and click Install.
5. After installation, click **Create HyperConverged** to create the deployment.
6. Accept all defaults and wait for status to show **Available**.

Step 2: Create a Virtual Machine

1. Navigate to **OpenShift Virtualization** in the left pane (refresh if needed).
2. Click **Create VirtualMachine**.
3. Choose a boot source (e.g., CentOS Stream 9 container disk).
4. Select a **General Purpose** instance type.
5. Under **Virtual Machine Details**, select the **fusionaccessfs** StorageClass from the dropdown.
6. Click **Create VirtualMachine**.

The VM will be created and started (takes a few minutes).

Step 3: Verify PVC Creation

Navigate to **Storage → PersistentVolumeClaims** in the OpenShift Virtualization namespace.

A PVC is created and bound using the `fusionaccessfs` storage class.

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```
oc get pvc -n openshift-cnv
```

Expected output:

```
NAME                    STATUS   VOLUME                                     CAPACITY
ACCESS MODES   STORAGECLASS         AGE
centos-stream9-dv       Bound    pvc-xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx   30Gi       RWX
{storage-class-name}   5m
```

Fileset Creation

IBM Storage Scale CNSA creates a **fileset** for each PVC. Filesets are recommended for: * Granular snapshots * Quota enforcement

Verify on the Scale cluster:

```
Fileset created for PVC: centos-stream9-dv
```

Step 4: Access the Virtual Machine

1. Navigate back to **OpenShift Virtualization** → **Virtual Machines**.
2. Wait for the VM to show **Running** state.
3. Click the VM name, then click **Console** to launch the VNC/Serial console.
4. Login with credentials displayed on the console screen.

Step 5: Generate I/O and Monitor Performance

Generate Write Traffic

In the VM console:

```
dd if=/dev/zero of=/mnt/testfile bs=1G count=5 oflag=direct
```

Monitor on IBM Storage Scale CNSA GUI

1. In Fusion Console, navigate to **Storage** → **Fusion Access for SAN**.
2. Click **Link to file system dashboard** to open the CNSA GUI.
3. Navigate to **Statistics** to view client I/O metrics for the filesystem.

You should see read/write throughput and IOPS for the VM workload.

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Step 6: Resiliency Test (Optional)

Simulate a storage node failure:

1. Cordon and drain a worker node that is part of the storage cluster but NOT running the VM (e.g., `worker-3`):

```
oc adm cordon {worker-node-prefix}3
oc adm drain {worker-node-prefix}3 --ignore-daemonsets --delete-emptydir-data
# Reboot the node via cloud provider console
```

1. Verify the VM continues running and generating I/O on the remaining nodes.
2. Check IBM Storage Scale CNSA GUI → Statistics to confirm continued I/O.
3. On the Scale node, verify sessions:

```
targetcli sessions
```

1. When the node returns, uncordon and verify it rejoins:

```
oc adm uncordon {worker-node-prefix}3
oc get pods -n ibm-spectrum-scale
targetcli sessions
```

All 3 workers should show `LOGGED_IN` and core pods should be running on the recovered node.

Step 7: Containerized Application PVC (Optional)

While Fusion Access for SAN is intended for OpenShift Virtualization, the underlying CNSA can also serve containerized applications.

1. Navigate to **Storage** → **PersistentVolumeClaims** in the `ibm-spectrum-scale` namespace.
2. Click **Create PersistentVolumeClaim**.
3. Configure:
   - Name: `cnsa-csi-pvc`
   - Size: `10GB`
   - StorageClass: `fusionaccessfs`
   - Access Mode: `RWX`
4. Click **Create** and monitor events.

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5. Verify the PVC is bound and available on the Scale cluster.

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Benchmarking IBM Fusion Access for SAN Storage

Introduction

This section demonstrates how to benchmark your IBM Fusion Access for SAN storage class using KubeStr.

For comprehensive storage benchmarking on Red Hat OpenShift with FIO, refer to: "Benchmarking Your Storage on Red Hat OpenShift with FIO" — IBM Community Guide

Tool: KubeStr

KubeStr is a Kubernetes storage benchmarking tool that can identify CSI drivers and run performance tests.

Download: KubeStr Releases

Step 1: Initial KubeStr Run — Identify CSI Drivers

Run KubeStr to discover all CSI drivers and their associated storage classes:

```
./kubestr fio
```

Expected output: KubeStr identifies the `fusionaccessfs` storage class and correctly identifies the CSI provider as **IBM Storage Scale**.

Step 2: Run Benchmark Against Fusion Access for SAN

Run KubeStr specifically against the `fusionaccessfs` storage class:

```
./kubestr fio --storage-class {storage-class-name}
```

Important Notes

Note

The lab environment is **not representative of a real-world deployment**: * Single-node IBM Storage Scale cluster * Nested virtualization * Single 10GbE network interface * iSCSI over single path (no multipath)

Benchmark results in this environment are for demonstration purposes only and do not reflect production performance.

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Production Benchmarking Considerations

For meaningful production benchmarks:

- **Bare-metal worker nodes** with dedicated SAN connectivity
- **Multiple Storage Scale nodes** for HA and performance
- **Multipath configuration** with multiple target portals
- **Appropriate workload profiles** (FIO job files matching your application I/O patterns)
- **Sufficient test duration** to reach steady state

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Summary & Licensing

Summary

This guide has demonstrated how to deploy **IBM Fusion Access for SAN** using iSCSI-attached storage, including:

1. **iSCSI Enablement on OpenShift RHCOS Worker Nodes**
   - Applied Machine Config to enable `iscsid`, `iscsi`, and `multipathd` services
   - Obtained IQNs for each worker node
   - Enabled multipath and verified configuration
2. **Linux-IO (LIO) iSCSI Target Configuration on IBM Storage Scale**
   - Installed `targetcli` and `iscsi-initiator-utils`
   - Created 500G backing store files on the Scale filesystem (`fs1`)
   - Configured iSCSI target IQN: `iqn.2022-06.com.scale.scalenode1:tgt0`
   - Created target portal group (TPG) on port 3260
   - Defined 2 LUNs backed by the fileio backing stores
   - Created ACLs for each OpenShift worker node IQN (no authentication for lab)
   - Saved configuration for persistence
3. **IBM Fusion Access for SAN Deployment**
   - Installed IBM Fusion Operator from IBM Operator Catalog
   - Deployed Fusion Data Foundation (FDF) on infra nodes for local storage
   - Configured OpenShift Image Registry storage
   - Enabled **External Systems → Storage Area Network** service
   - Selected worker nodes for the storage cluster
   - Created filesystem `fusionaccessfs` backed by iSCSI LUNs
   - Verified CNSA pod deployment and filesystem health
   - Confirmed automatic StorageClass creation (`fusionaccessfs`)
4. **Virtual Machine Deployment**

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- Installed OpenShift Virtualization Operator
- Created HyperConverged deployment
- Deployed VM using `fusionaccessfs` StorageClass
- Verified PVC creation and fileset on Scale cluster
- Generated I/O and monitored performance via CNSA GUI

5. **Resiliency Validation**
   - Simulated worker node failure (cordon, drain, reboot)
   - Verified VM continued running with I/O on remaining nodes
   - Confirmed node recovery and rejoin to Scale cluster
6. **Containerized Application Support (Optional)**
   - Demonstrated PVC creation for containerized workloads using `fusionaccessfs`
   - Verified fileset creation on Scale cluster

Key Takeaways

- **Storage Agnostic:** IBM Fusion Access for SAN works with any backend storage vendor
- **Built on Proven Technology:** Leverages IBM Storage Scale's parallel architecture and resiliency
- **Low-Latency, High-Performance:** Delivers block storage performance over FC or iSCSI
- **Simplified Management:** OpenShift-native UI for storage cluster, filesystem, and StorageClass management
- **Enterprise Ready:** GA since IBM Fusion 2.11 / OpenShift 4.19 (August 2025)

Licensing & Quotes

IBM Fusion Access for SAN licensing information and quote requests:

- Licensing portal: IBM Fusion Access Licensing

Appendix A: targetcli Configuration Reference

The following is the targetcli configuration file (`/etc/target/saveconfig.json`) created in this guide. Use as a reference to validate your configuration. The full file is available at `examples/yaml/targetcli-saveconfig.json`.

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```
{
  "fabric_modules": [],
  "storage_objects": [
    {
      "aio": false,
      "alua_tpgs": [
        {
          "alua_access_state": 0,
          "alua_access_status": 0,
          "alua_access_type": 3,
          "alua_support_active_nonoptimized": 1,
          "alua_support_active_optimized": 1,
          "alua_support_offline": 1,
          "alua_support_standby": 1,
          "alua_support_transitioning": 1,
          "alua_support_unavailable": 1,
          "alua_write_metadata": 0,
          "implicit_trans_secs": 0,
          "name": "default_tg_pt_gp",
          "nonop_delay_msecs": 100,
          "preferred": 0,
          "tg_pt_gp_id": 0,
          "trans_delay_msecs": 0
        }
      ],
      "attributes": {
        "alua_support": 1,
        "block_size": 512,
        "emulate_3pc": 1,
        "emulate_caw": 1,
        "emulate_dpo": 1,
        "emulate_fua_read": 1,
        "emulate_fua_write": 1,
        "emulate_model_alias": 0,
        "emulate_pr": 1,
        "emulate_rest_reord": 0,
        "emulate_tas": 1,
        "emulate_tpu": 0,
        "emulate_tpws": 0,
        "emulate_ua_intlck_ctrl": 0,
        "emulate_write_cache": 0,
        "enforce_pr_isids": 1,
```


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```
        "force_pr_aptpl": 0,
        "is_nonrot": 0,
        "max_unmap_block_desc_count": 1,
        "max_unmap_lba_count": 8192,
        "max_write_same_len": 4096,
        "optimal_sectors": 16384,
        "pgr_support": 1,
        "pi_prot_format": 0,
        "pi_prot_type": 0,
        "pi_prot_verify": 0,
        "queue_depth": 128,
        "unmap_granularity": 1,
        "unmap_granularity_alignment": 0,
        "unmap_zeroes_data": 0
      },
      "dev": "/scale/fs1/iscsilun1",
      "name": "scale-fs1-iscsilun1",
      "plugin": "fileio",
      "size": 536870912000,
      "write_back": false,
      "wwn": "781b293e-fbf0-49fd-9b45-9a18347b49de"
    },
    {
      "aio": false,
      "alua_tpgs": [
        {
          "alua_access_state": 0,
          "alua_access_status": 0,
          "alua_access_type": 3,
          "alua_support_active_nonoptimized": 1,
          "alua_support_active_optimized": 1,
          "alua_support_offline": 1,
          "alua_support_standby": 1,
          "alua_support_transitioning": 1,
          "alua_support_unavailable": 1,
          "alua_write_metadata": 0,
          "implicit_trans_secs": 0,
          "name": "default_tg_pt_gp",
          "nonop_delay_msecs": 100,
          "preferred": 0,
          "tg_pt_gp_id": 0,
          "trans_delay_msecs": 0
```


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```
        }
      ],
      "attributes": {
        "alua_support": 1,
        "block_size": 512,
        "emulate_3pc": 1,
        "emulate_caw": 1,
        "emulate_dpo": 1,
        "emulate_fua_read": 1,
        "emulate_fua_write": 1,
        "emulate_model_alias": 0,
        "emulate_pr": 1,
        "emulate_rest_reord": 0,
        "emulate_tas": 1,
        "emulate_tpu": 0,
        "emulate_tpws": 0,
        "emulate_ua_intlck_ctrl": 0,
        "emulate_write_cache": 0,
        "enforce_pr_isids": 1,
        "force_pr_aptpl": 0,
        "is_nonrot": 0,
        "max_unmap_block_desc_count": 1,
        "max_unmap_lba_count": 8192,
        "max_write_same_len": 4096,
        "optimal_sectors": 16384,
        "pgr_support": 1,
        "pi_prot_format": 0,
        "pi_prot_type": 0,
        "pi_prot_verify": 0,
        "queue_depth": 128,
        "unmap_granularity": 1,
        "unmap_granularity_alignment": 0,
        "unmap_zeroes_data": 0
      },
      "dev": "/scale/fs1/iscsilun0",
      "name": "scale-fs1-iscsilun0",
      "plugin": "fileio",
      "size": 536870912000,
      "write_back": true,
      "wwn": "3248cc42-749c-4b5e-93eb-6b0065c89d45"
    }
  ],
```


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```
  "targets": [
    {
      "attributes": {
        "authentication": -1,
        "dataout_timeout": 3,
        "dataout_timeout_retries": 5,
        "default_erl": 0,
        "nopin_response_timeout": 30,
        "nopin_timeout": 15,
        "random_datain_pdu_offsets": 0,
        "random_datain_seq_offsets": 0,
        "random_r2t_offsets": 0
      },
      "node_acls": [
        {
          "attributes": {
            "authentication": -1,
            "dataout_timeout": 3,
            "dataout_timeout_retries": 5,
            "default_erl": 0,
            "nopin_response_timeout": 30,
            "nopin_timeout": 15,
            "random_datain_pdu_offsets": 0,
            "random_datain_seq_offsets": 0,
            "random_r2t_offsets": 0
          },
          "mapped_luns": [
            {
              "alias": "cec2b1fdc4",
              "index": 0,
              "tpg_lun": 0,
              "write_protect": false
            },
            {
              "alias": "eae96eaffe",
              "index": 1,
              "tpg_lun": 1,
              "write_protect": false
            }
          ]
        }
      ],
```

```
      "portals": [
        {
          "ip_address": "0.0.0.0",
          "iser": false,
          "offload": false,
          "port": 3260
        }
      ],
      "tag": 1,
      "wwn": "iqn.2022-06.com.scale.scalenode1:tgt0"
    }
  ]
}
```


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